BIOMECHANICS (EPHE 341)

Dynamic Analysis

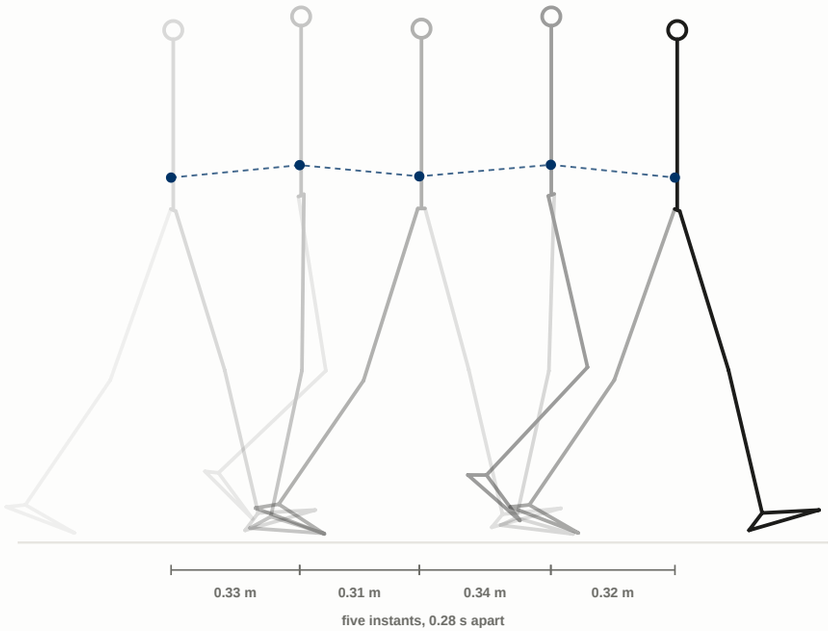
Last week every equation had zero on the right-hand side. This week the zeros become *ma* and *Iα*, and the method does not change.

Dynamic Analysis

Dynamic Equilibrium

- State of an object which is moving with constant* (uniform) linear and/or angular velocity. * no acceleration.
- Dynamic equilibrium exists if there is a balance between the applied forces and inertial forces for a body in motion, with all the applied forces resulting in equal and oppositely directed inertial forces.
- Therefore the same rules apply from static to dynamic equilibrium but you need to keep in mind that the object is in motion.

> Equal times, equal distances. That is the whole test, and you can run it on any recording you have.

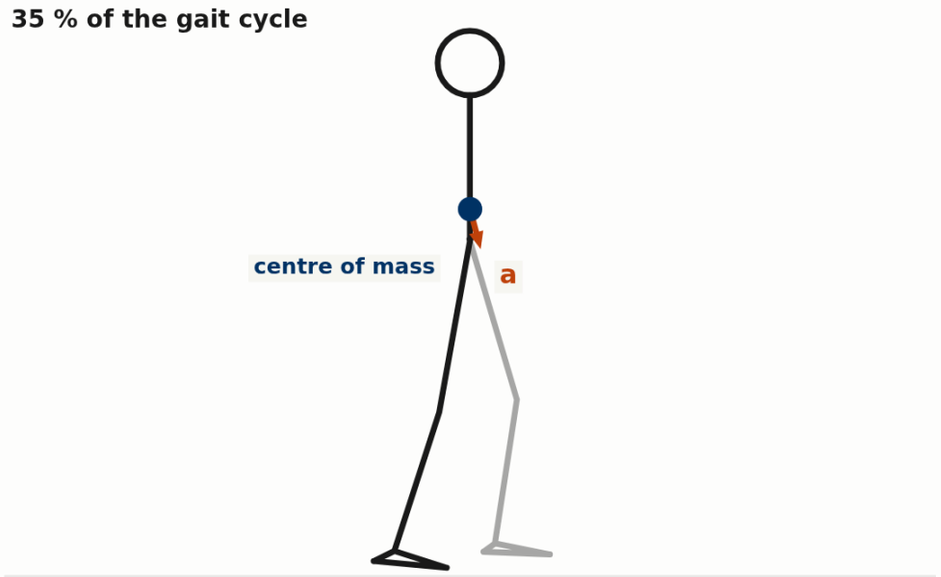


One stride of a real walk, five instants 0.28 s apart. The dot is the whole-body centre of mass.

Dynamic Analysis

When it is not in equilibrium

When an object is **not** in equilibrium the net forces do not equal zero and it requires a non-equilibrium **dynamic analysis**. To understand dynamic analysis we need to start with Newton's second law.



forward speed of the centre of mass

1.09 m/s

mean over the stride 1.15, and it never stops changing

forward acceleration

0.23 m/s²

if this were equilibrium it would be zero

vertical acceleration

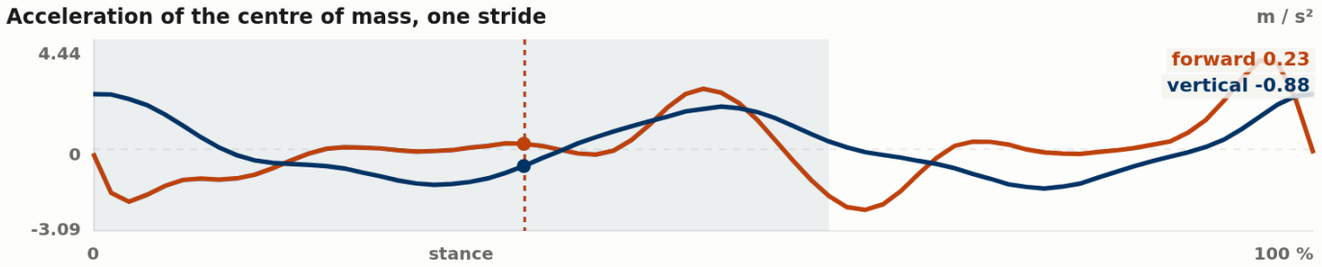
−0.88 m/s²

the centre of mass rises and falls 34 mm, twice a stride

so the net force is

13 N across, 491 N up

55 kg × a, and the weight is 540 N



Where in the stride

35 %

▶ Walk

The same stride as the last slide, differentiated. The centre of mass accelerates by up to 4.4 m/s² forwards and 2.7 m/s² vertically, so **ΣF is not zero** and the static method does not apply. Nothing here uses the force plate — the markers alone are enough to show a body is not in equilibrium.

Dynamic Analysis

Newton's Second Law — the law of acceleration

LINEAR

A net force applied to a body causes an acceleration of that body of a magnitude proportional to the force, in the direction of the force, and inversely proportional to the body's mass.

$$\Sigma F = ma$$

ANGULAR

A net torque (moment) applied to a body causes an angular acceleration of that body of a magnitude proportional to the torque, in the direction of the torque (moment), and inversely proportional to the moment of inertia.

$$\Sigma M = I\alpha$$

> Two cautions that cost marks later. The *a* in *F* = *ma* is the acceleration of the **centre of mass** — not of the heel, not of the joint. And the *M* and the *I* in *M* = *I*α must be taken about the **same axis**.

Dynamic Analysis

In two dimensions

- In a 2-dimensional analysis it is important to consider the components of forces and moment(s).
- This leaves us with the **3 equations** of a system under dynamic conditions.

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma M = I\alpha$$



Jakob Emanuel Handmann, 1753

Leonhard Euler (1707–1783) wrote the rotational law for a rigid body, and the two laws together describe any motion of anything solid. Hence the **Newton–Euler** equations, which is what this method is called in every paper that uses it.

> Three equations means **three unknowns**, and not one more. Everything about how inverse dynamics is organised follows from that one sentence.

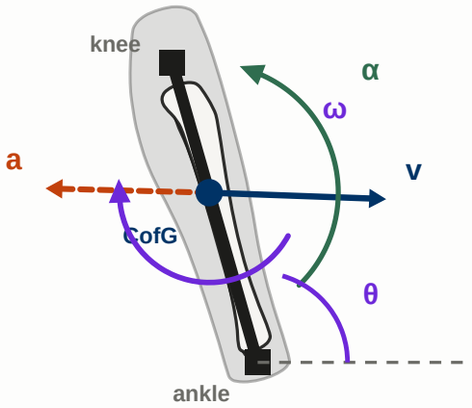
Dynamic Analysis

Why two dimensions

- Dynamic analysis of a multi-segment rigid body (like a human body) in 3-dimensions can be very complex.
- We will cover **2-dimensional** examples of dynamic analysis.

In the plane a segment can do exactly three things: move across, move up, and turn. **Three degrees of freedom, three equations** — the same three from the last slide. In three dimensions it is six of each.

> This is not only a teaching simplification. A large fraction of published gait analysis reports sagittal joint moments computed exactly this way, because the sagittal plane is where most of walking happens.



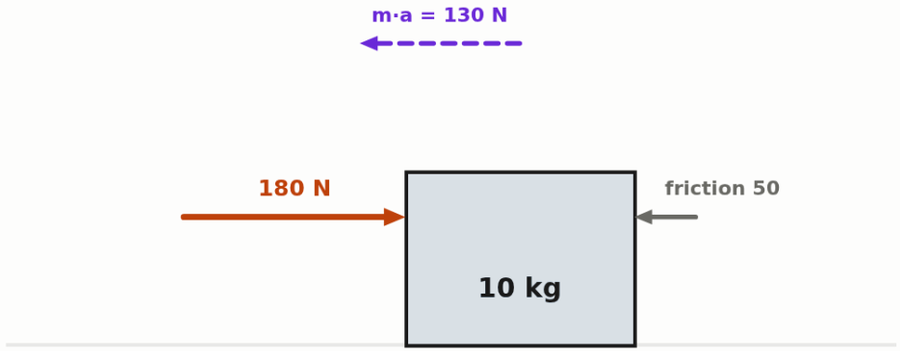
θ	74°	where it is pointing
ω	+2.74 rad/s	how fast it is turning
α	-66.8 rad/s²	and how that changes
v	(+1.62, -0.06) m/s	the centre of gravity, moving
a	(-7.62, +0.21) m/s²	and accelerating

One shank, 90 % of the way through a measured stride. Blue and orange belong to the centre of gravity; violet and green belong to the segment.

Dynamic Analysis

The same free body, with and without acceleration

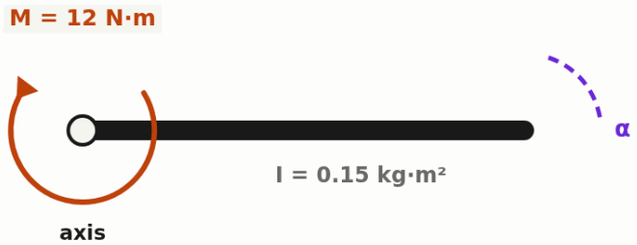
ΣFx = m·ax



180 − 50 = 10 · a

a = 13.00 m/s²

ΣM = I·α



12 = 0.15 · α

α = 80.0 rad/s²

The free body diagram does not change · the force table does not change · the sign conventions do not change · the only thing that changes is that the right-hand side is no longer zero

The push **180 N**

The twist **12 N·m**

Last week the right-hand side was zero and you solved for the forces that made it so. This week it is **m·a** and **I·α**, and you solve for the forces that produce the motion you measured. Slide the push down to 50 N, where it exactly cancels friction, and the dynamic problem turns back into the static one without a single line of the method changing. The dashed arrow is the quantity m·a drawn as though it were a force: add it and the sums are zero again, which is why we call this **dynamic equilibrium**.

Two directions

Forces in, motion out — or motion in, forces out. One of them you can do to a person.

Dynamic Analysis

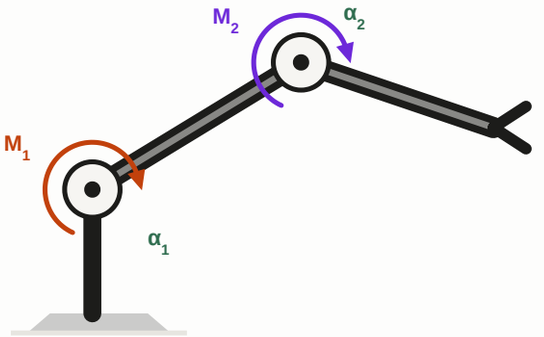
Forward dynamics

- There are **2 approaches** to solving dynamic problems depending on what variables are known or can be manipulated.
- Forward and inverse dynamics.

FORWARD DYNAMICS ANALYSIS

- When you know the forces and are trying to determine the accelerations.
- An example of this is in robotics where you have motors that produce a prescribed force or torque.
- That is, given the forces and torques, what are the linear and angular accelerations?

$$\text{forces} \rightarrow a, \alpha \rightarrow \text{motion}$$



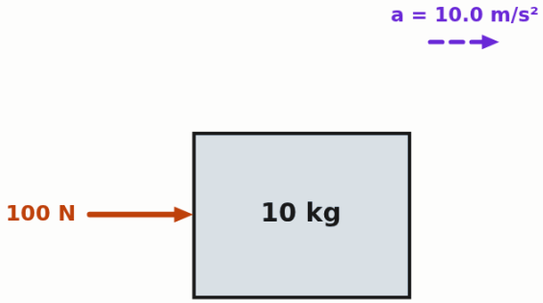
Each motor is told what torque to produce. Everything the arm then does follows from /α.

This is the direction every physics problem you have ever been set runs in. It is also the direction you **cannot** run on a person, because nobody can tell you what the muscle forces were.

Dynamic Analysis

Example Problem #1 (forward dynamics)

What is the acceleration if a force of 100 N is applied to an object with a mass of 10 kg?



A force of 100 N on a 10 kg object. What is the acceleration?

ΣF = m·a

the forces all act through the centre of gravity, so nothing turns

ΣF = 100 = 100 N

a = 100 / 10 = 10.00 m/s²

the slide too

Forward dynamics: the forces are given and the motion comes out. This is the direction a simulation runs, and the direction you can almost never use on a person, because the forces inside are the part you cannot measure.

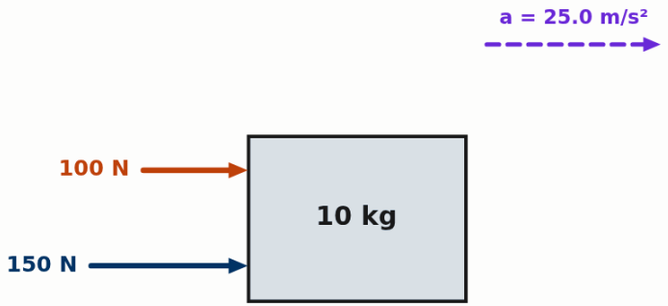
The first force **100 N**

Forward dynamics: **known forces in, motion out**. Move the slider and watch the answer follow it in proportion — that is the whole of the second law. Nothing here needs a free body diagram with unknowns in it, which is exactly what makes the inverse problem — the rest of this lecture — harder.

Dynamic Analysis

Example Problem #2 (forward dynamics)

What is the acceleration if a force of 100 N and a force of 150 N are applied to a 10 kg object such that no rotation can occur — that is, both forces pass through the centre of gravity?



100 N and 150 N, both through the centre of gravity, on 10 kg.

ΣF = m·a

the forces all act through the centre of gravity, so nothing turns

ΣF = 100 + 150 = 250 N

a = 250 / 10 = 25.00 m/s²

the slide too

Forward dynamics: the forces are given and the motion comes out. This is the direction a simulation runs, and the direction you can almost never use on a person, because the forces inside are the part you cannot measure.

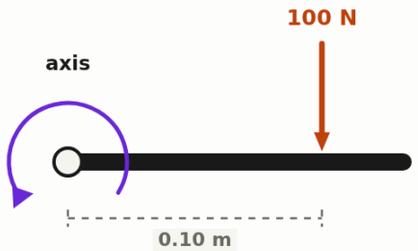
The first force **100 N**

Forward dynamics: **known forces in, motion out**. Move the slider and watch the answer follow it in proportion — that is the whole of the second law. Nothing here needs a free body diagram with unknowns in it, which is exactly what makes the inverse problem — the rest of this lecture — harder.

Dynamic Analysis

Example Problem #3 (forward dynamics)

Calculate the angular acceleration caused by the 100 N force on the segment if the moment of inertia of the segment is 0.15 kg·m².



I = 0.15 kg·m²

100 N at 0.1 m from the axis, on a segment of I = 0.15 kg·m².

ΣM = I·α

each force is perpendicular to the segment, so the moment arm is the distance itself

ΣM = 100(0.10) = 10.0 N·m

α = 10.0 / 0.15 = 66.67 rad/s²

the slide too

The caption under this slide reads "this clockwise torque will cause a negative moment and angular acceleration" — and it will, so the answer is −66.67 rad/s². The printed answer has lost its sign. In a lecture whose worked example turns on sign conventions that is worth saying out loud.

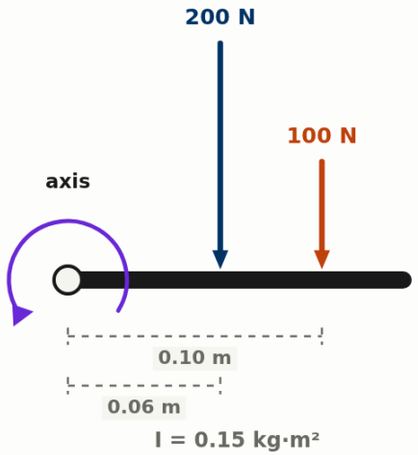
The first force — **100 N**

Forward dynamics: **known forces in, motion out**. Move the slider and watch the answer follow it in proportion — that is the whole of the second law. Nothing here needs a free body diagram with unknowns in it, which is exactly what makes the inverse problem — the rest of this lecture — harder.

Dynamic Analysis

Example Problem #4 (forward dynamics)

Calculate the angular acceleration caused by both the 100 N and the 200 N forces on the segment if the moment of inertia of the segment is 0.15 kg·m².



100 N at 0.1 m and 200 N at 0.06 m, on I = 0.15 kg·m².

ΣM = I·α

each force is perpendicular to the segment, so the moment arm is the distance itself

ΣM = 100(0.10) + 200(0.06) = 22.0 N·m

α = 22.0 / 0.15 = 146.67 rad/s²

the slide too

The caption under this slide reads "this clockwise torque will cause a negative moment and angular acceleration" — and it will, so the answer is −146.67 rad/s². The printed answer has lost its sign. In a lecture whose worked example turns on sign conventions that is worth saying out loud.

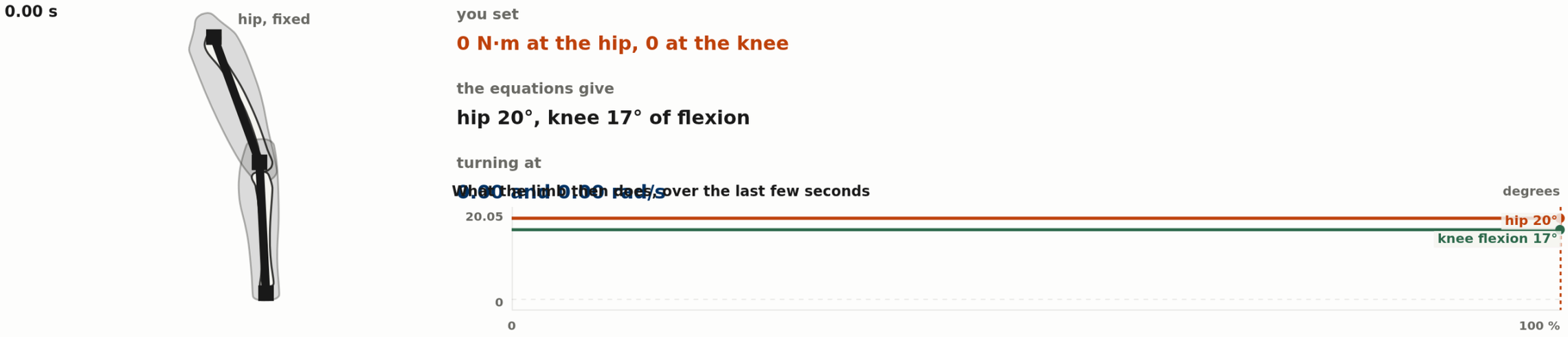
The first force — **100 N**

Forward dynamics: **known forces in, motion out**. Move the slider and watch the answer follow it in proportion — that is the whole of the second law. Nothing here needs a free body diagram with unknowns in it, which is exactly what makes the inverse problem — the rest of this lecture — harder.

Dynamic Analysis

Simulation

Another application of forward dynamics is in the **simulation of human movement**, when approximate muscle forces are input into models to replicate the performance of a movement.



Forward dynamics, run forwards. You set two torques; everything else is the equations of motion integrated in time. Masses, lengths and inertias are the same subject's, off the same table, and each joint carries a passive stiffness, a damper and its own end stops. **Nothing here is a recording** — change a torque and the movement changes, which is what you cannot do to a person.

Dynamic Analysis

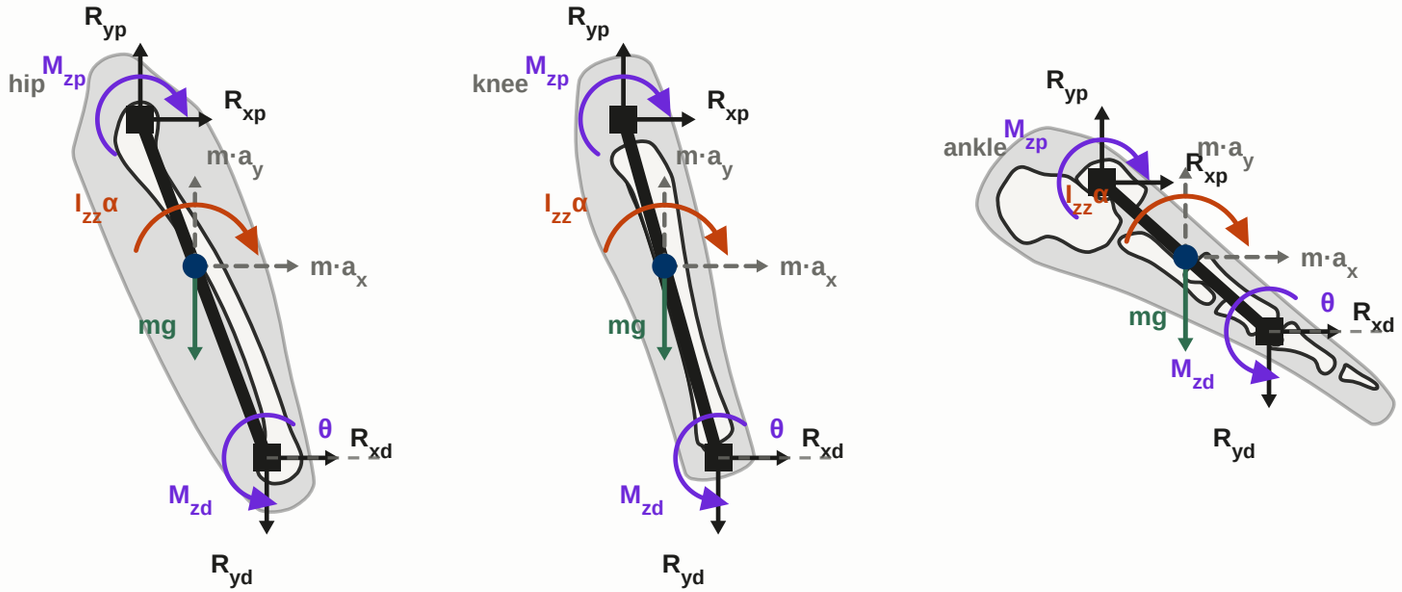
Inverse dynamics analysis

- When you know the **kinematics** and are trying to solve for the **kinetics**.
- An example of this type of solution is when you can measure the movement of the body segment and want to solve for the muscle force and joint reaction forces required to produce that movement.

$$\text{motion} + \text{external forces} \rightarrow \text{internal forces}$$

> You cannot put a load cell inside somebody. You can put a camera in front of them and a plate under them — so this is the only direction available.

Everything from here on is this one calculation, and the figure is what it must account for: **ten per segment**, thirty for the limb.

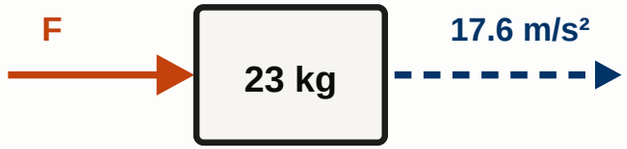


Each cut carries R_x, R_y and M_z; each segment carries mg, ma_x, ma_y, $I\alpha$ and θ.

Dynamic Analysis

Example Problem #1 (inverse dynamics)

What is the resultant force applied to a 23 kg object if the acceleration is 17.6 m/s²?



SOLUTION

$$\Sigma F_x = m \cdot a_x$$

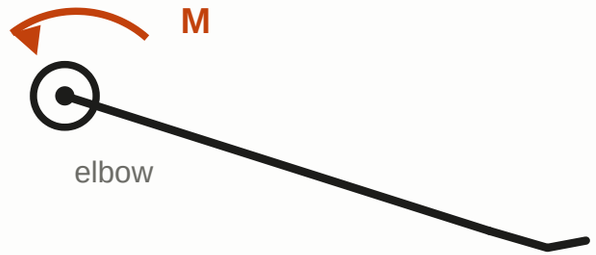
$$F = 23(17.6)$$

$$F = 404.8 \text{ N}$$

Dynamic Analysis

Example Problem #1 (inverse dynamics)

Calculate the moment required for an angular acceleration of 112 rad/s², given the moment of inertia about the elbow is 0.18 kg·m².



SOLUTION

$$\Sigma M = I \cdot \alpha$$

$$M = 0.18(112)$$

$$M = 20.2 \text{ N·m}$$

The method

Break the limb into segments, start at the one touching the ground, and carry the answer upward.

Inverse Dynamics

Inverse dynamics in human gait analysis

- We can record the movement of the human body using a motion capture system, video camera, accelerometers, gyros, etc…
- and measure the external forces,
- but we do not know what is going on **inside** the body to produce the motion.

> Based on recorded data (motion trajectories, GRFs) and simple dynamics principles (Newton–Euler equations) we can reconstruct the **net joint moments and forces**.

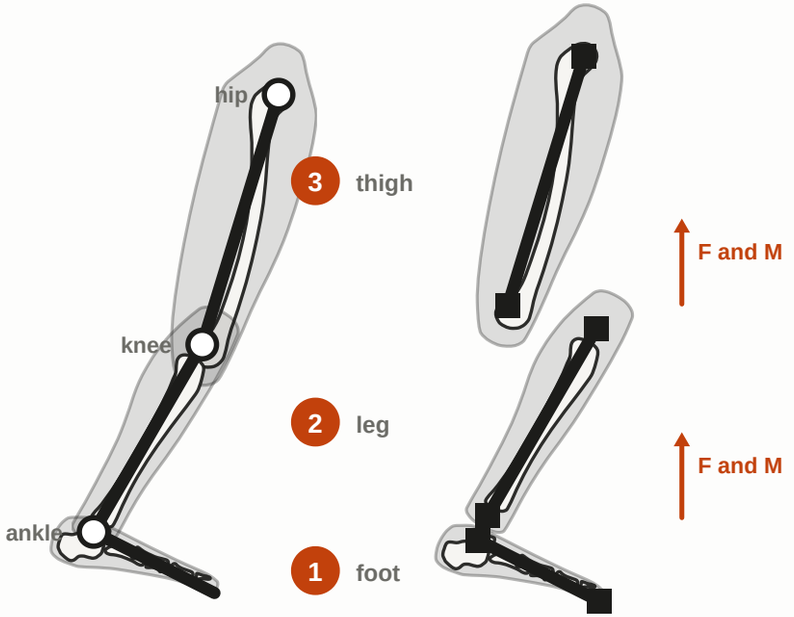
The word **net** is doing a great deal of work in that sentence, and we come back to it before the end.

Inverse Dynamics

One segment at a time, starting at the bottom

- Inverse dynamics involves taking into account all the forces and moments that act on each segment, and the effect of these forces and moments on the following segments.
- For example, if we want to find out the forces and moments at the hip during a movement we need to consider **each segment separately**.
- We need to start with the **most distal segment** and then carry the solutions for each joint forward.

Step 1: Foot · **Step 2:** Leg · **Step 3:** Thigh

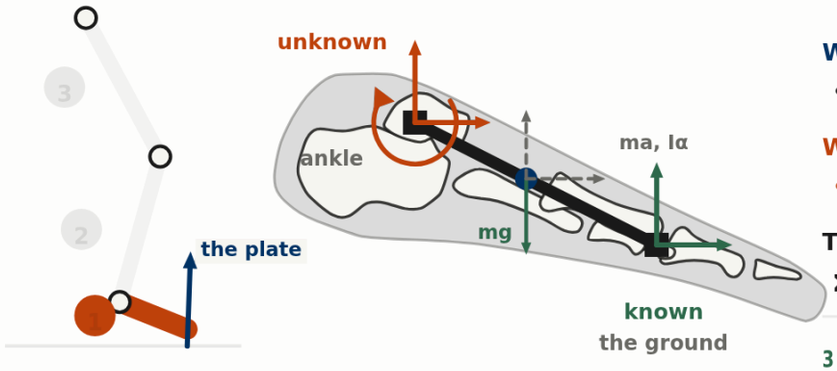


drawn at one instant of the recording — 47 % of the stride, heel off

Inverse Dynamics

Why the order is forced

Step 1 · the foot



What arrives at the distal end

- **GRFx, GRFy — measured by the plate**
- **where it acts — the centre of pressure**
- **no moment: the ground pushes, it does not twist**

What the anthropometric table gives

- **m, I and the centre of mass of the foot, and a and α from the markers**

What is unknown — three things

- **Fankle-x Fankle-y Mankle**

Three equations close it

ΣFx = max ΣFy = may ΣM = Iα

3 unknowns, 3 equations — solvable. Then the ankle force and the ankle moment go up to step 2.

A free body in two dimensions gives three equations. An unknown joint costs three unknowns. So a segment can be solved only when one of its ends is already known — and at the start exactly one end of one segment is.

Step 1 · the foot | Step 2 · the leg | Step 3 · the thigh | **hide the limb**

Three segments, three solutions, each one handing the next the force and the moment it needs. The **foot** can go first because a force plate has already measured what happens at its far end, and because the ground applies no moment. Everything above it is the same three equations with the previous answer arriving, reversed, at the bottom.

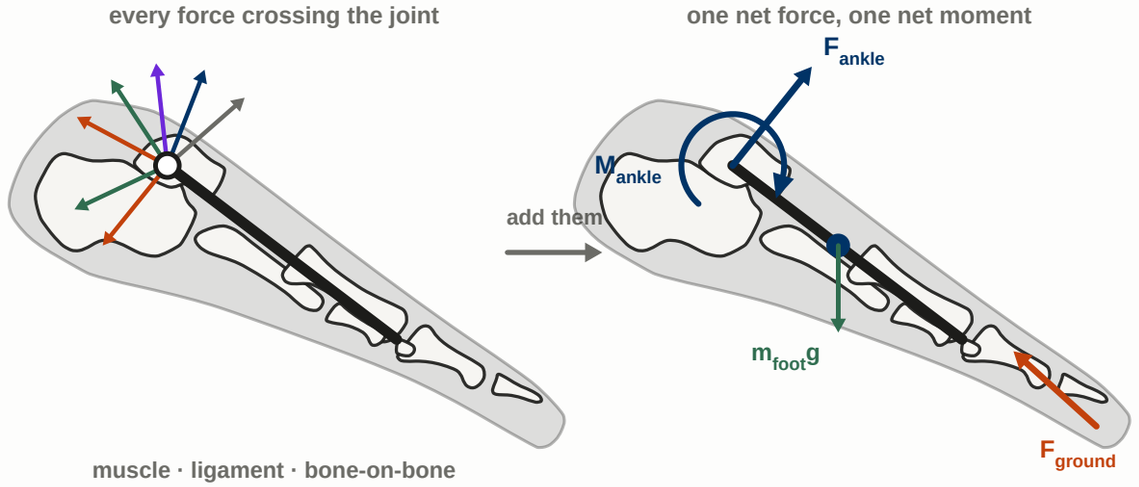
Inverse Dynamics

A net joint force and a net joint moment

1. We need to consider all the forces acting upon the segment and the moments that they would produce — bone-on-bone forces, ligament forces, every muscle crossing the joint and the joint moment of force it produces.

2. Really we can say that we can add all these forces to create a **net joint force** and a **net joint moment**, and then account for the internal and external moments.

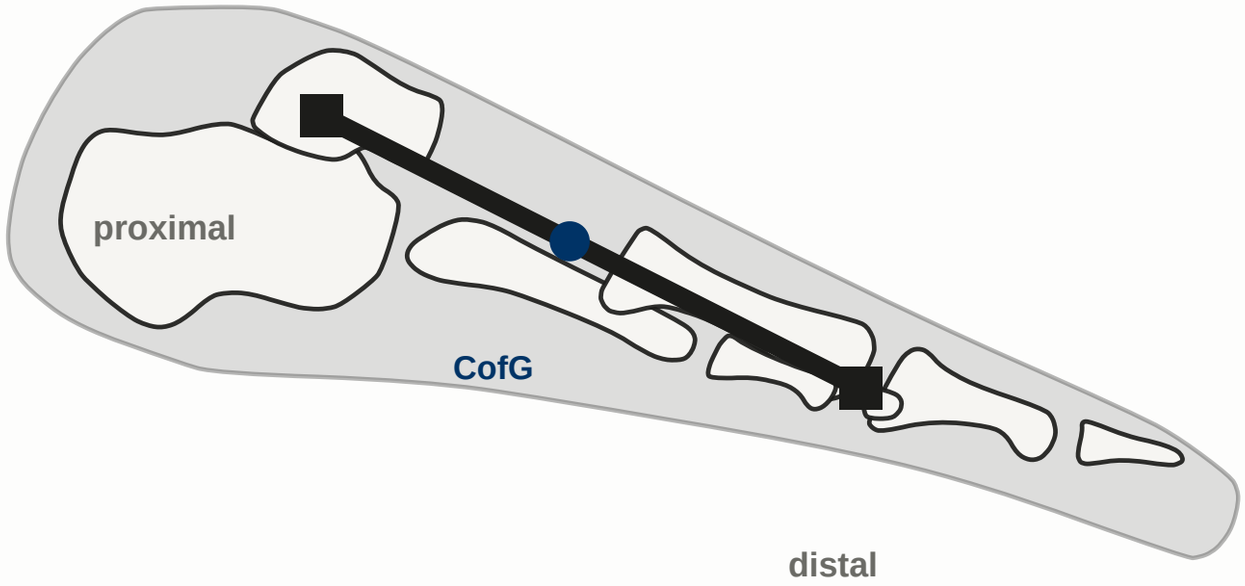
> One force and one moment, at **each joint, for each segment**. That replacement is exact for the mechanics — the segment accelerates identically — and lossy for the anatomy, because afterwards there is no way of telling which muscle did it.



Inverse Dynamics

A really simple diagram

So let's make a really simple diagram so we can see the steps necessary to calculate the net joint forces and moments.



$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma M = I\alpha$$

NB. The technique for calculating the net joint forces and moments is slightly different from the technique we are used to. It is the same in principle, but it may feel a bit different because we are taking moments about a different axis. Either technique could be used, but the one I am showing you simplifies the calculations given the number of variables.

Inverse Dynamics

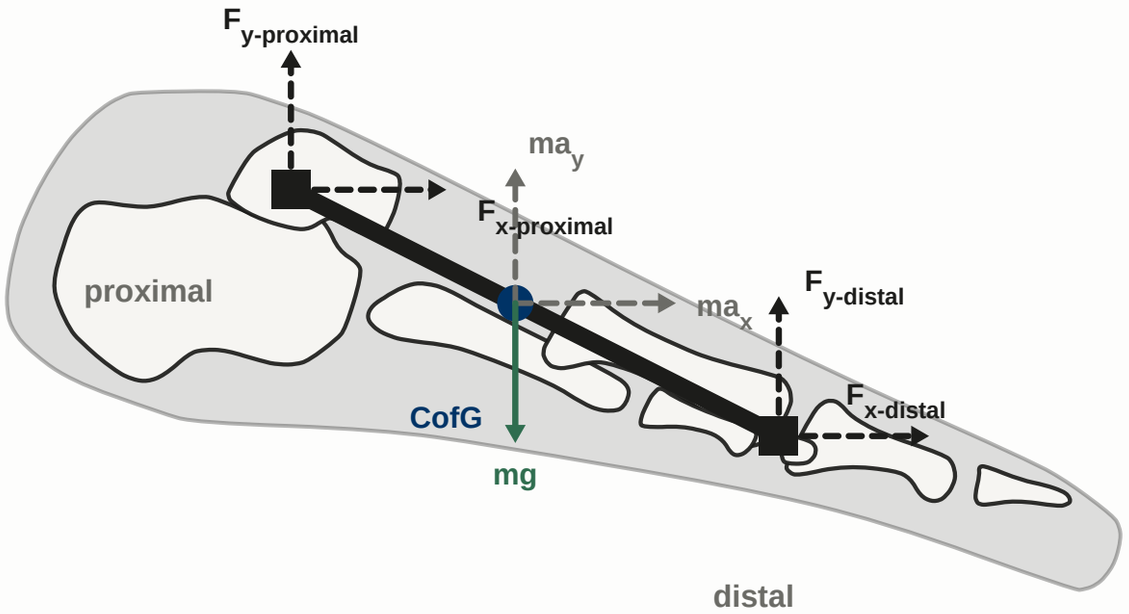
The forces, in general form

If we calculate all values with reference to the **centre of mass** we can use the general forms of the Newton–Euler equations that we learned earlier.

$$\Sigma F_{\mathrm{x}} = F_{\text{x-proximal}} + F_{\text{x-distal}} = ma_{\mathrm{x}}$$

$$\Sigma F_{\mathrm{y}} = F_{\text{y-proximal}} + F_{\text{y-distal}} + mg = ma_{\mathrm{y}}$$

> Hover any term to find it on the diagram. And while these are the general forms of the equations you need to keep close attention to the **directions (signs)** of the forces when calculating the values.



The forces about the **distal** end are the ground reaction forces. The forces about the **proximal** end are normally the unknown forces.

Inverse Dynamics

The forces, with specific values

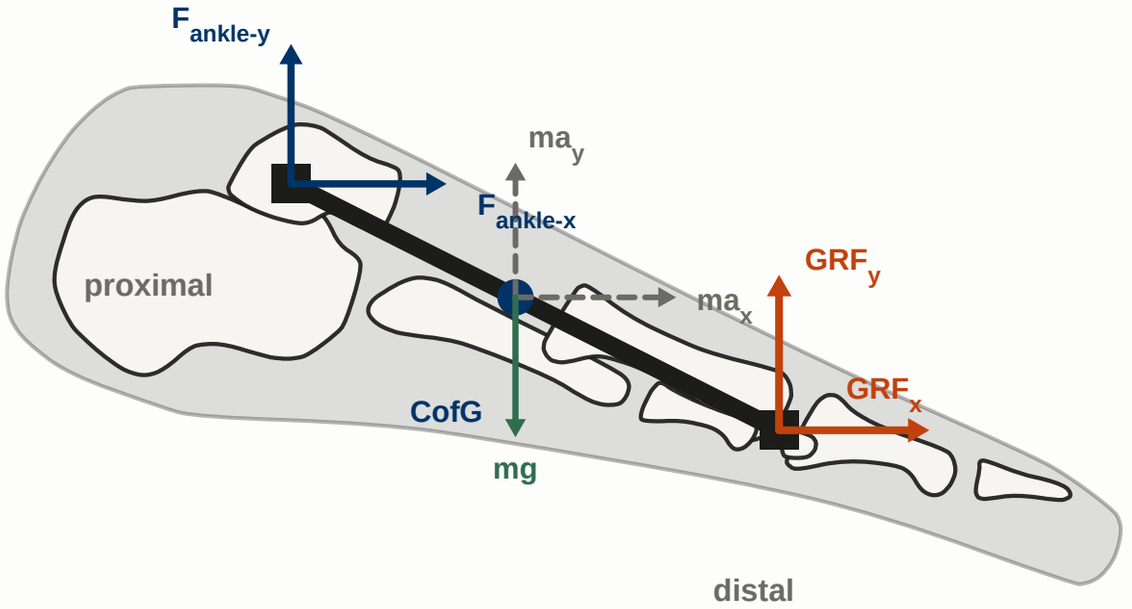
In this example we can give specific forces.

$$\Sigma F_{\mathrm{x}} = F_{\text{ankle-x}} + \mathrm{GRF}_{\mathrm{x}} = ma_{\mathrm{x}}$$

$$F_{\text{ankle-x}} = ma_{\mathrm{x}} - \mathrm{GRF}_{\mathrm{x}}$$

$$\Sigma F_{\mathrm{y}} = F_{\text{ankle-y}} + \mathrm{GRF}_{\mathrm{y}} + mg = ma_{\mathrm{y}}$$

$$F_{\text{ankle-y}} = ma_{\mathrm{y}} - \mathrm{GRF}_{\mathrm{y}} - mg$$



The muscle force acting at the ankle · the ground reaction force

Inverse Dynamics

The moments

$$\Sigma M = I\alpha$$

$$M_{\text{ankle}} + M_{\text{Fankle/CG}} + M_{\text{GRF/CG}} = I\alpha$$

M_{ankle}

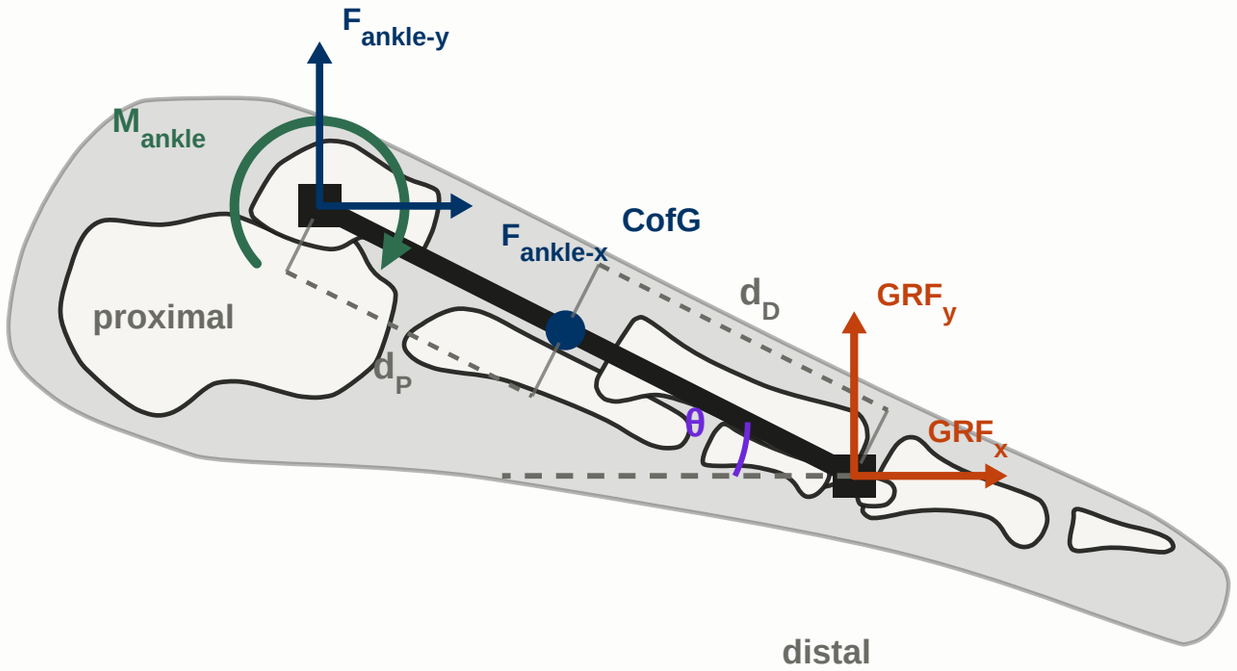
$+ F_{\text{ankle-x}} \cdot d_P(\sin\theta)$

$+ F_{\text{ankle-y}} \cdot d_P(\cos\theta)$

$+ \text{GRF}_x \cdot d_D(\sin\theta)$

$+ \text{GRF}_y \cdot d_D(\cos\theta)$

$= I\alpha$



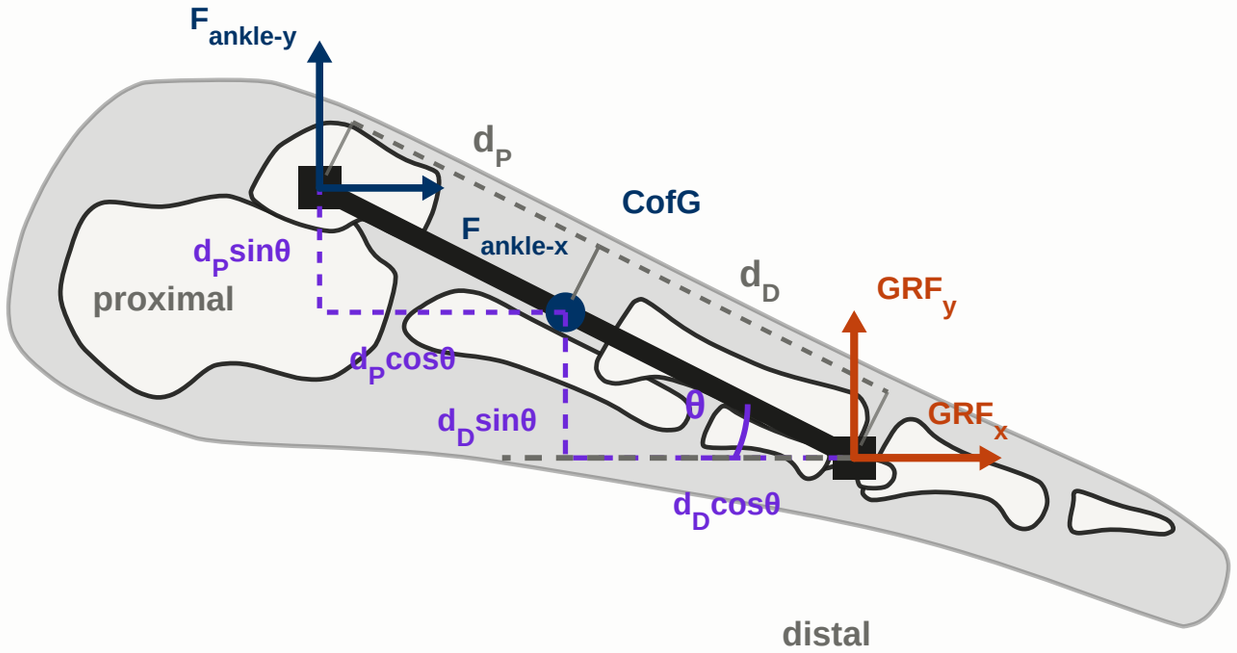
Inverse Dynamics

The same moments, with the geometry drawn

$$\Sigma M = I\alpha$$

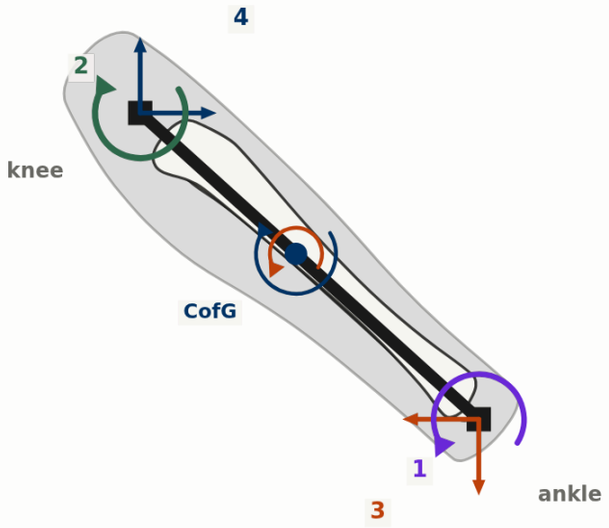
M_{ankle}
$+ F_{ankle\text{-}x} \cdot d_P(\sin\theta)$
$+ F_{ankle\text{-}y} \cdot d_P(\cos\theta)$
$+ GRF_x \cdot d_D(\sin\theta)$
$+ GRF_y \cdot d_D(\cos\theta)$
$= I\alpha$

$d_P\sin\theta$ and $d_D\sin\theta$ are **vertical** offsets, and they go with the **horizontal** forces. The arm a force turns through is perpendicular to it.



Inverse Dynamics

Every segment carries four moments



Four moments, and they add to Iα

4	**−M ankle** the moment applied at the far end	**−0.394 N·m**
2	**M knee** the moment at the joint we are solving	**1.716 N·m** the unknown
3	**M Fankle/CG** the far-end force, about the CofG	**−1.094 N·m**
1	**M Fknee/CG** the joint force, about the CofG	**2.141 N·m**
	sum of the four	**2.369 = Iα = 2.369 N·m**

Four terms, four moments. Term 1 came up from the foot; term 2 is what goes up to the thigh.

The foot, in swing | The foot, in stance | **The leg** | **hide the limb**

All four | 1 · moment at the far end | 2 · moment at the joint | 3 · far-end force about the CofG | 4 · joint force about the CofG

Every segment has **four** moments acting on it and they add to **Iα**: one at each end, and one about the centre of gravity from each end's force. **Term 2 is always the unknown** — the moment at the proximal end is what we are solving for, and it becomes term 1 of the segment above.

Inverse Dynamics

Up the chain

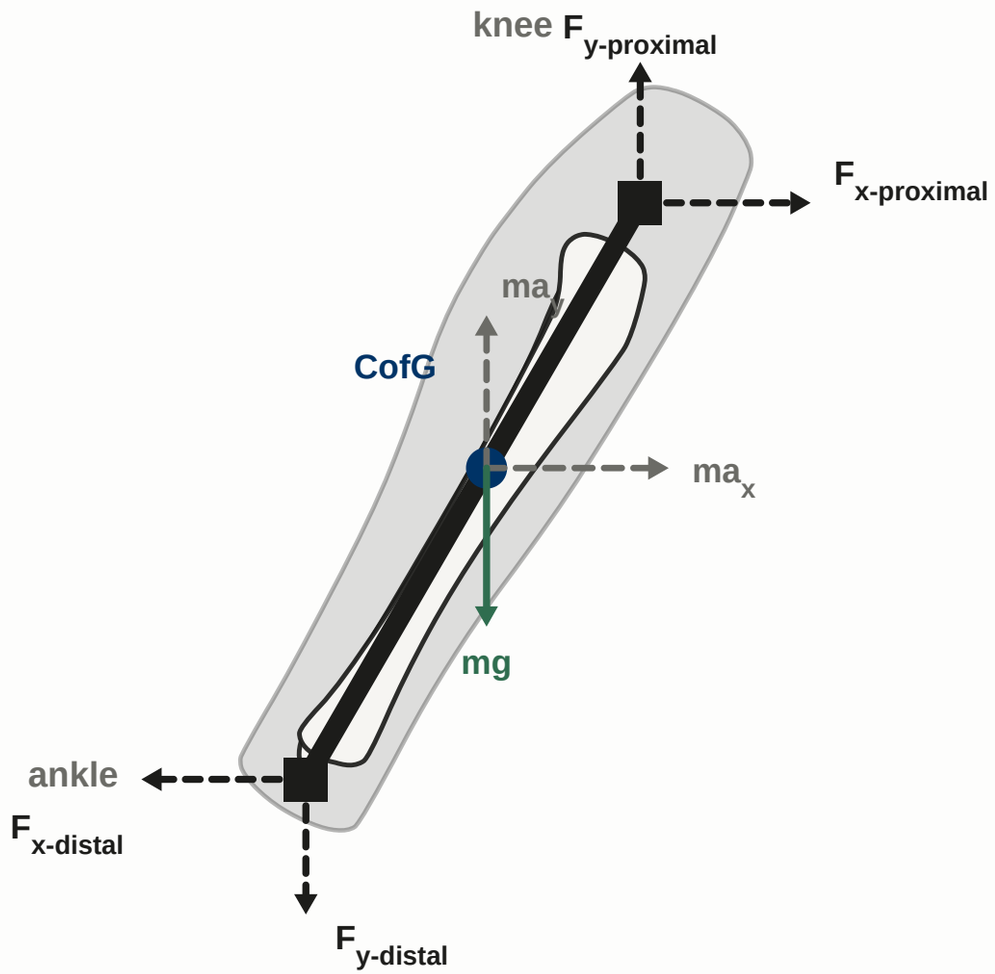
Now that we have found the net ankle joint force and moment we can then use that information to input into the **next limb segment**.

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma M = I\alpha$$

Same three equations, new segment. The leg's distal end is the ankle, and we have just finished measuring it.



The distal arrows point the other way now — that is Newton's third law, and it is the next slide.

Inverse Dynamics

The leg, with specific forces

In this example we can give specific forces.

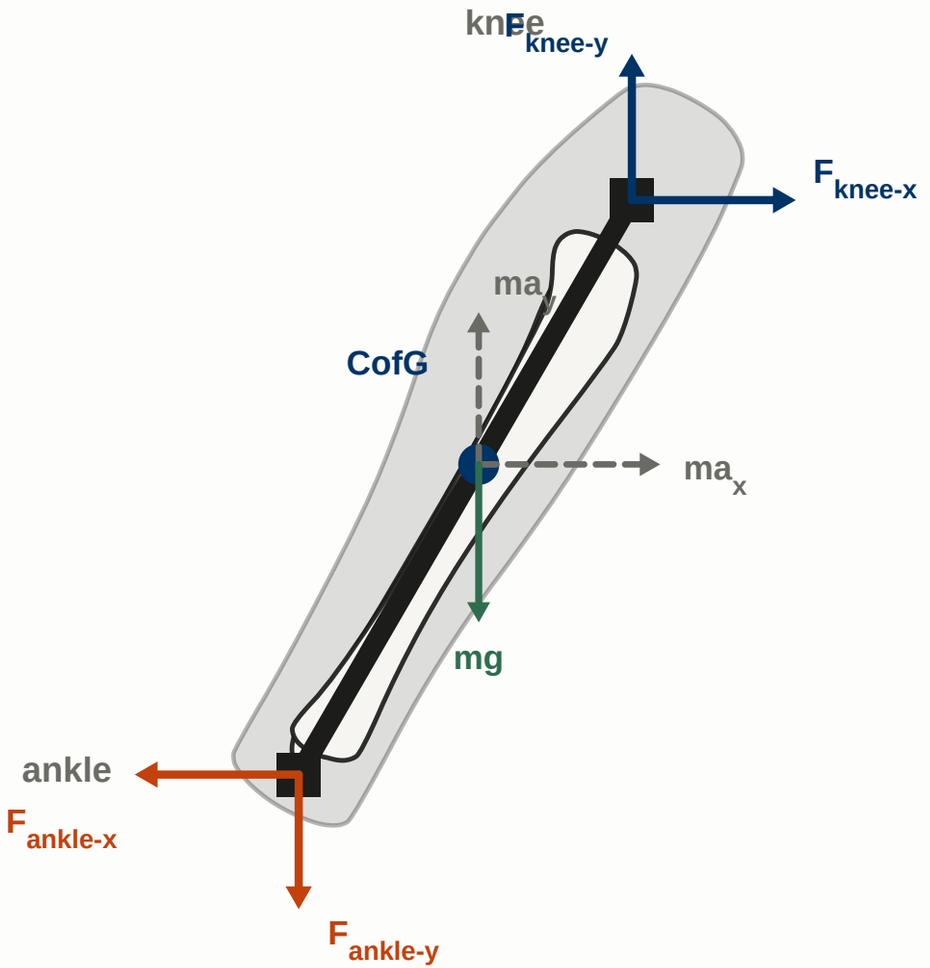
$$\Sigma F_{\text{x}} = F_{\text{knee-x}} - F_{\text{ankle-x}} = ma_{\text{x(leg)}}$$

$$F_{\text{knee-x}} = ma_{\text{x}} + F_{\text{ankle-x}}$$

$$\Sigma F_{\text{y}} = F_{\text{knee-y}} - F_{\text{ankle-y}} + mg = ma_{\text{y(leg)}}$$

$$F_{\text{knee-y}} = ma_{\text{y}} + F_{\text{ankle-y}} - mg$$

> Need to consider **action/reaction** forces. Those minus signs are the whole content of this slide.



The muscle force acting at the knee · the ankle reaction force

Inverse Dynamics

The leg, with specific moments

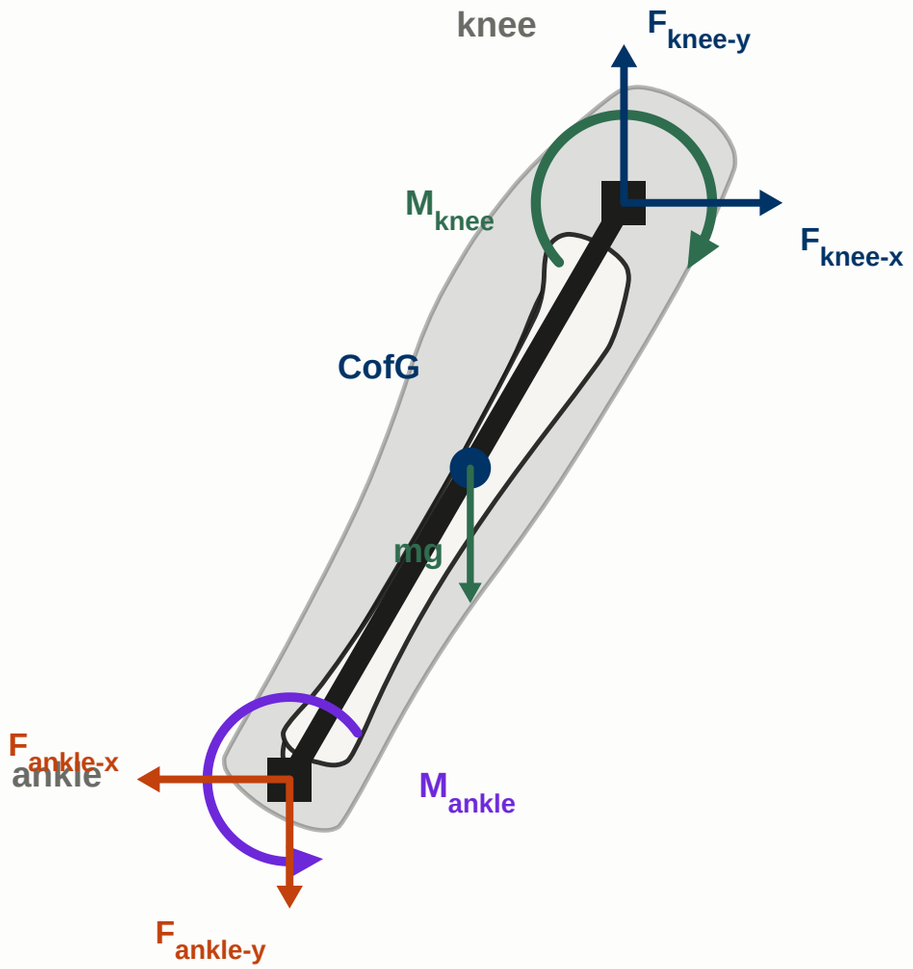
In this example we can give specific moments.

$$\Sigma M = I\alpha$$

$$M_{\text{knee}} + M_{\text{Fknee/CG}} + M_{\text{Fankle/CG}} - M_{\text{ankle}} = I\alpha$$

$$M_{\text{knee}} = I\alpha - M_{\text{Fknee/CG}} - M_{\text{Fankle/CG}} + M_{\text{ankle}}$$

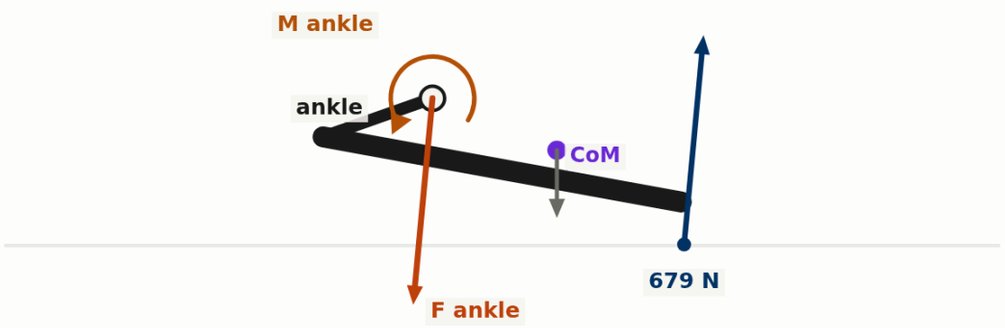
> Really just remember that what we solved for at the ankle will affect each limb up the chain.



A moment at **both** ends now — the one we want at the knee, and the one we already solved at the ankle.

Inverse Dynamics

The same free body, on a measured stride



44 % of the gait cycle

ΣFx = m·ax

F(ankle)x + 63 = 0.80(0.79)

F(ankle)x = −62.1 N

ΣFy = m·ay

F(ankle)y + 676 − 7.8 = 0.80(1.33)

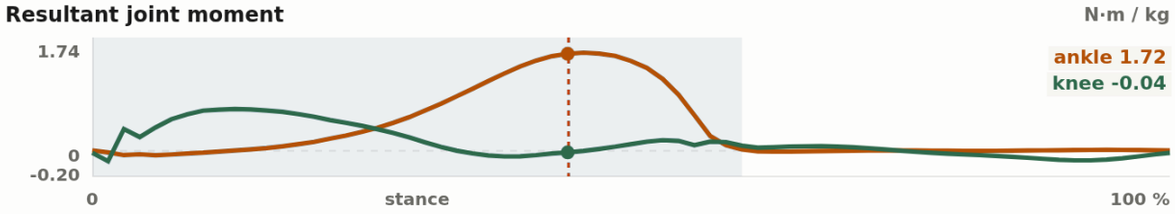
F(ankle)y = −667.2 N

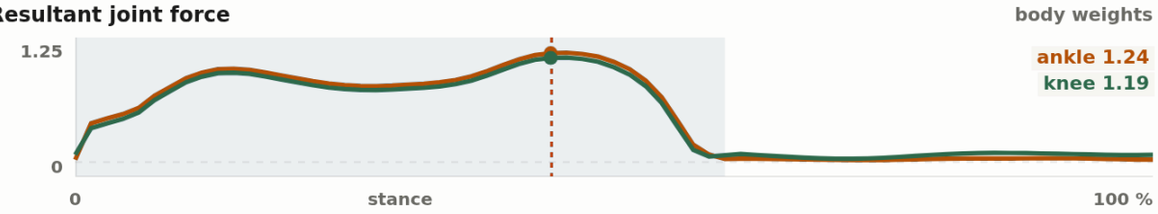
ΣM = I·α

M(ankle) + M(F ankle) + M(GRF) = 0.00381(−25.6)

M(ankle) = −94.48 N·m

that is 1.718 N·m per kilogram, counting a plantarflexor moment as positive





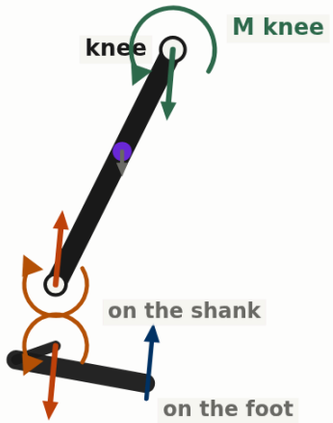
Where in the stride — 44 %

The diagram | The forces | **The moment** | **▶ Walk**

The free body from the last few slides, with a real foot in it, and both answers across the whole stride. Past 60 % the plate loses the foot and the ankle moment collapses to almost nothing.

Inverse Dynamics

Handing the ankle to the knee



44 % of the gait cycle

what the foot produced

F = (−62.1, −667.2) N, M = −94.48 N·m

reversed onto the shank

F = (62.1, 667.2) N, M = 94.48 N·m

ΣFx, ΣFy at the knee

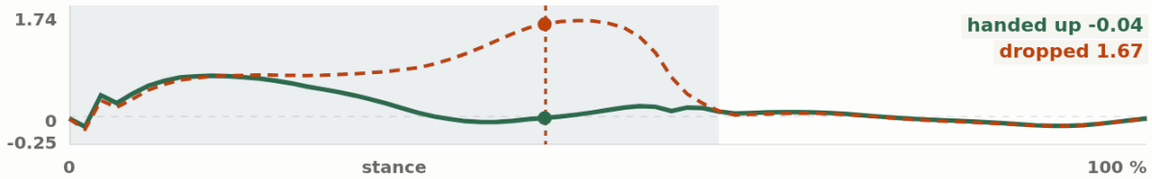
F(knee) = (−54.1, −638.2) N

ΣM = I·α at the knee

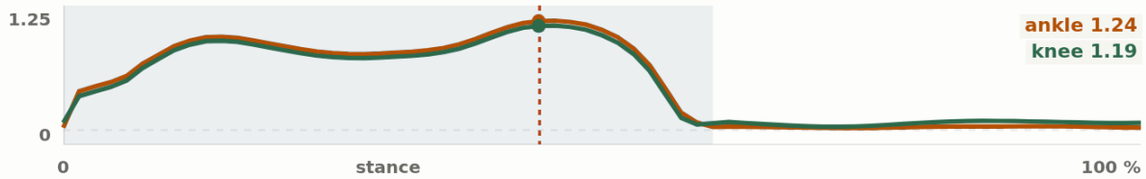
M(knee) = −2.40 N·m = −0.044 N·m/kg

Both halves of the handover are in. At this instant dropping the moment alone would move the knee answer by 94.48 N·m.

Knee moment, with the handover and without



Resultant joint force



Where in the stride — **44 %**

Hand the moment up | Forget the moment | **▶ Walk**

Newton’s third law applies to moments as well as to forces: the foot’s free body produced **two** things at the ankle and **both** come back reversed, as the terms **− M(ankle)** and **− F(ankle)**.

Inverse Dynamics

In summary…

What do you need to remember from the last few slides? In order to solve for joint forces and moments:

- You need to know the **kinematics** of the movement and any **external forces**.
- Break up each multi-limb segment into **individual limbs**.
- Use the **Newton–Euler equations** to solve for unknown net forces and moments for each joint, starting from the most distal segment.
- **Transfer** forces and moments up the link of segments.

This gives you an idea of the forces and moments experienced at each joint that result from the external forces and the movements.

Inverse Dynamics

What inverse dynamics can and cannot tell us

WHAT IT CAN TELL US

- **NET** joint moments, powers and forces.

WHAT IT CANNOT TELL US

- Contributions/activations of individual muscles.
- Total amount of muscle work done — it does not account for co-contraction of muscles.
- Internal bone-to-bone forces at joints.

OTHER ASSUMPTIONS AND LIMITATIONS

- Frictionless joints.
- Generalized, uniform and/or concentrated mass distribution.
- Estimation of joint centre of rotation.
- Measurement errors: skin and motion artifacts, temporal or spatial misalignment between force plates and motion capture.

The worked example

The ankle and the knee during the swing phase of walking, line by line.

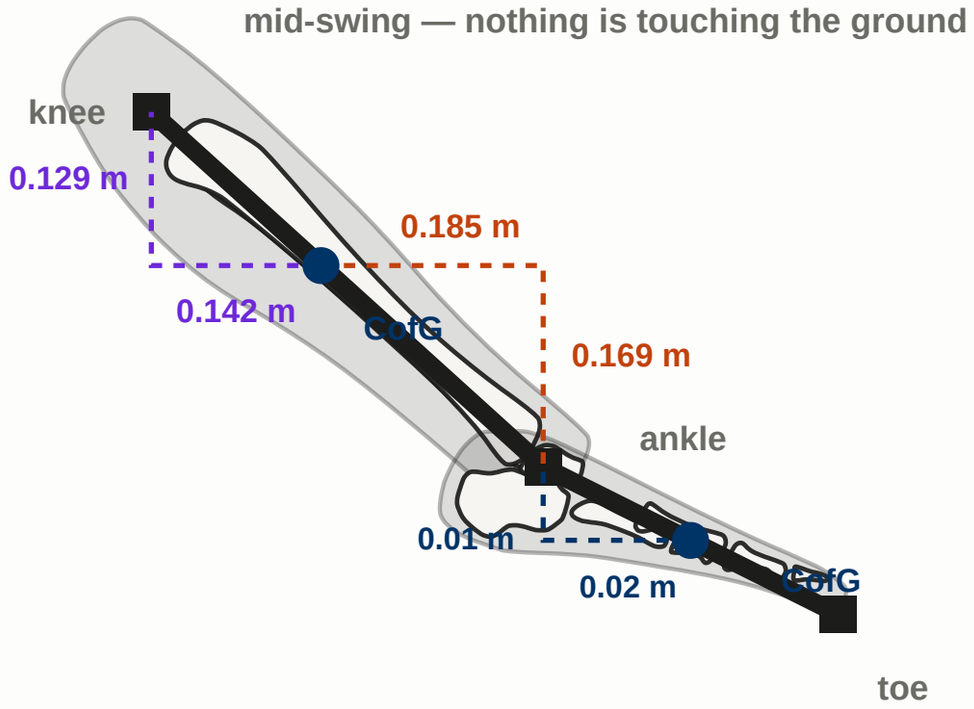
Inverse Dynamics Analysis Example

What you are given

Calculate the forces and moments about the ankle and knee during the **swing phase** of walking using the data below.

	Foot	Leg
m (kg)	1.16	3.72
I (kg·m²)	0.00995	0.0642
a_x (m/s²)	9.07	−0.03
a_y (m/s²)	−6.62	−4.21
α (rad/s²)	21.63	36.9

Where did those masses come from? Winter's table gives the foot 1.45 % of body mass and the leg 4.65 %. 1.16 / 0.0145 = **80.0 kg**. 3.72 / 0.0465 = **80.0 kg**. The same person, to three figures.



The distances from each joint to each segment's centre of mass come from the same table. Those are the moment arms.

Inverse Dynamics Analysis Example

The foot, the two force equations

In this example we can give specific forces. In swing there is no ground reaction force, so $F_{\text{x-distal}}$ and $F_{\text{y-distal}}$ are both zero.

$$\Sigma F_x = F_{\text{ankle-x}} + \text{GRF}_x = ma_x$$

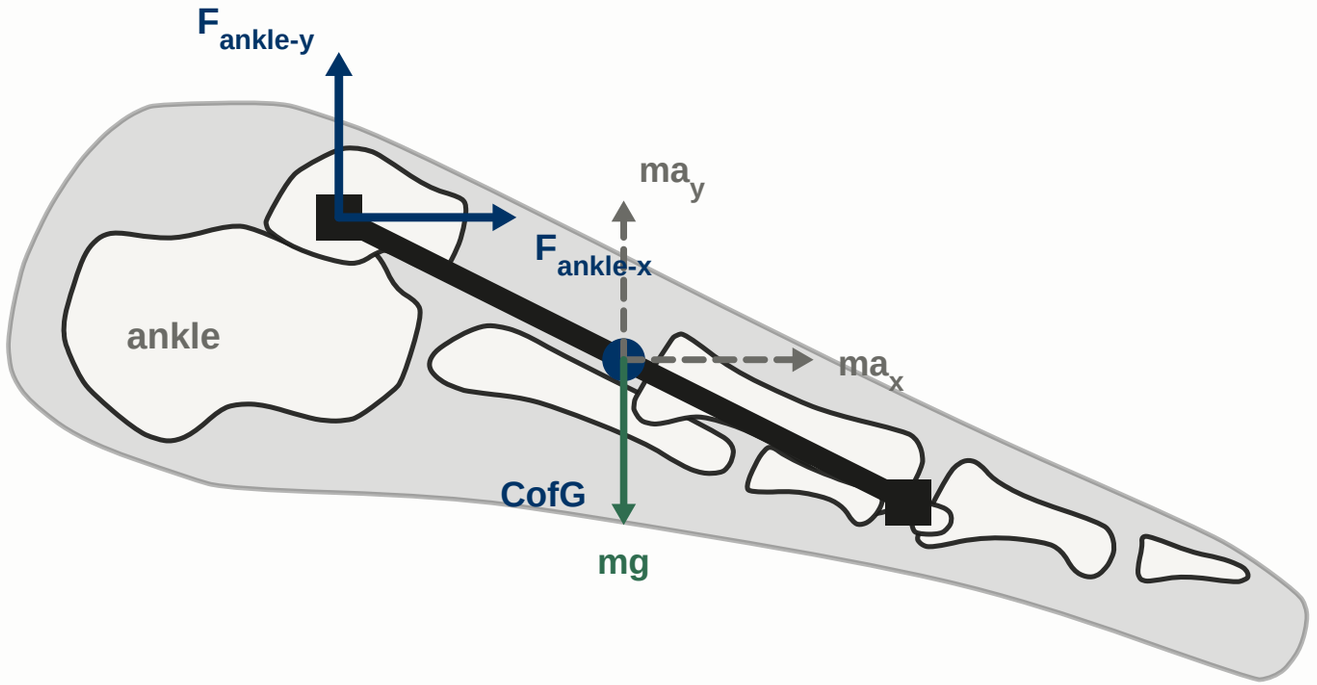
$$F_{\text{ankle-x}} = ma_x = (1.16\ \text{kg})(9.07\ \text{m/s}^2)$$

$$F_{\text{ankle-x}} = 10.52\ \text{N}$$

$$\Sigma F_y = F_{\text{ankle-y}} + \text{GRF}_y + mg = ma_y$$

$$F_{\text{ankle-y}} = ma_y - mg = (1.16)(-6.62) - (1.16)(-9.81)$$

$$F_{\text{ankle-y}} = -7.68 + 11.38 = 3.70\ \text{N}$$



It is very important to pay close attention to the **signs**.

Inverse Dynamics Analysis Example

The foot, setting up the moment equation

$$\Sigma M = I\alpha$$

$$M_{\text{ankle}} + M_{\text{Fankle/CG}} + M_{\text{GRF/CG}} = I\alpha$$

M_{ankle}

$+ F_{\text{ankle-x}} \cdot d_P(\sin\theta)$

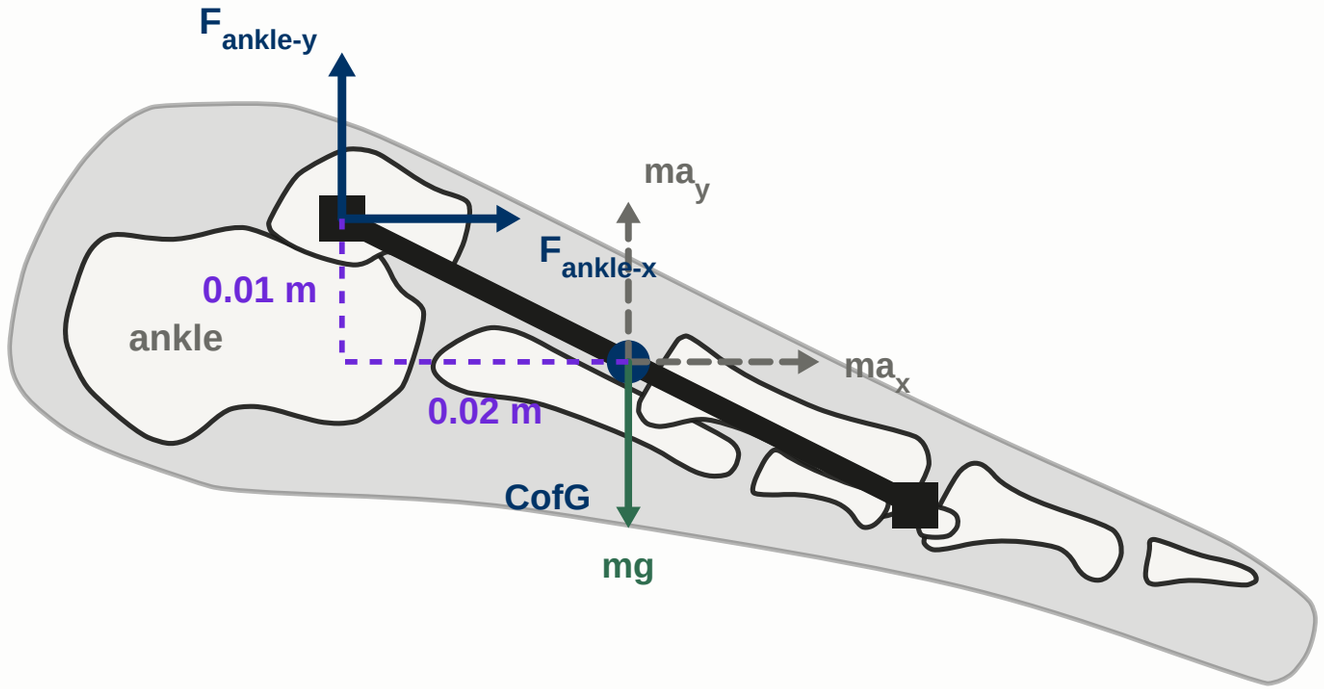
$+ F_{\text{ankle-y}} \cdot d_P(\cos\theta)$

$+ \text{GRF}_x \cdot d_D(\sin\theta) = 0$

$+ \text{GRF}_y \cdot d_D(\cos\theta) = 0$

$= I\alpha$

> Because this is during swing there are **no moments caused by the GRF**. Two of the five terms are gone before any arithmetic starts.



$d_P\sin\theta$ is the **0.01 m** vertical offset and goes with the horizontal force; $d_P\cos\theta$ is the **0.02 m** horizontal one.

Inverse Dynamics Analysis Example

The foot, the ankle moment

$$M_{\text{ankle}} + F_{\text{ankle-x}}d_{\text{Px}} + F_{\text{ankle-y}}d_{\text{Py}} = I\alpha$$

Based on the **direction** of the forces you can determine whether they will cause a positive or a negative moment. Therefore:

$$M_{\text{ankle}} - F_{\text{ankle-x}}d_{\text{Px}} - F_{\text{ankle-y}}d_{\text{Py}} = I\alpha$$

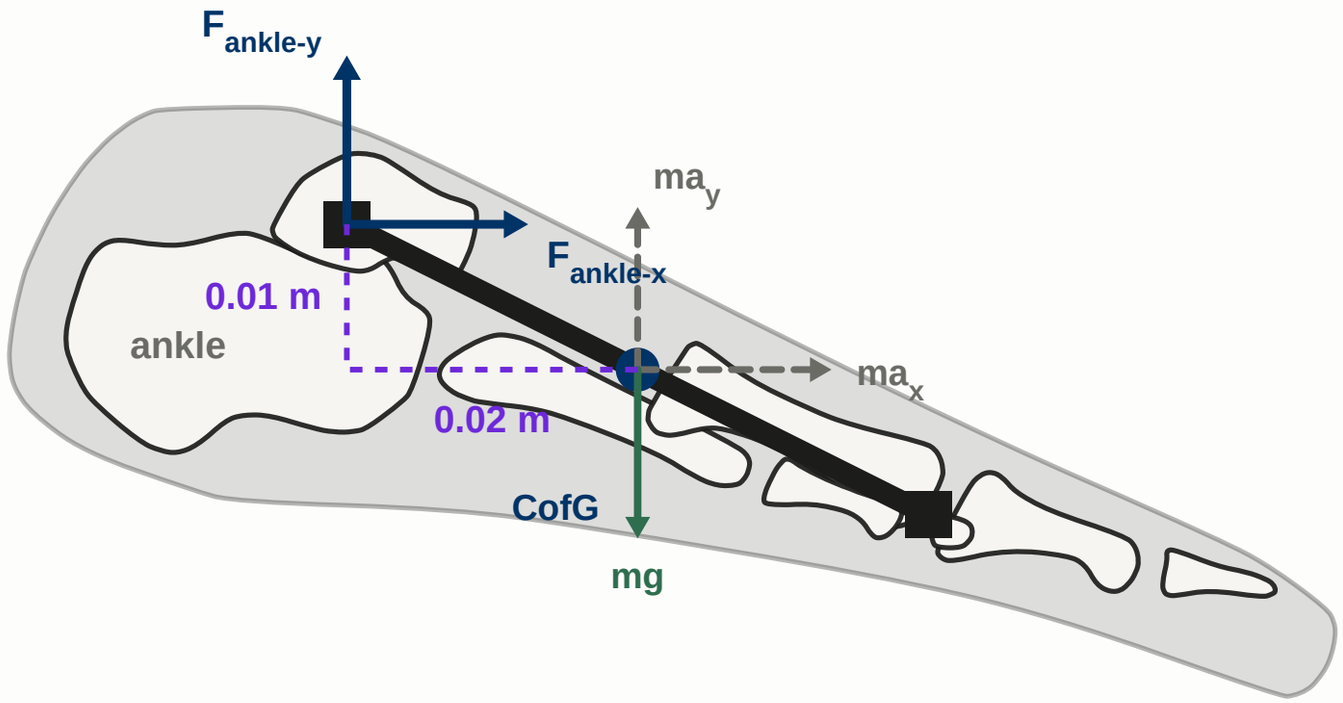
$$M_{\text{ankle}} = F_{\text{ankle-x}}d_{\text{Px}} + F_{\text{ankle-y}}d_{\text{Py}} + I\alpha$$

$$M_{\text{ankle}} = (10.52\ \text{N})(0.01\ \text{m}) + (3.70\ \text{N})(0.02\ \text{m}) + (0.00995\ \text{kg·m}^2)(21.63\ \text{rad/s}^2)$$

$$M_{\text{ankle}} = 0.105\ \text{N·m} + 0.074\ \text{N·m} + 0.2152\ \text{N·m}$$

$$M_{\text{ankle}} = 0.39\ \text{N·m}$$

This positive torque denotes an ankle **dorsiflexor** moment.



Three terms, and the largest of them is *Iα* — the foot is being swung, not loaded.

Inverse Dynamics Analysis Example

The leg, the two force equations

Need to consider **action/reaction** forces: what we found at the ankle arrives at the leg reversed.

$$\Sigma F_x = F_{\text{knee-x}} - F_{\text{ankle-x}} = ma_{x(\text{leg})}$$

$$F_{\text{knee-x}} - 10.52\ \text{N} = (3.72\ \text{kg})(-0.03\ \text{m/s}^2)$$

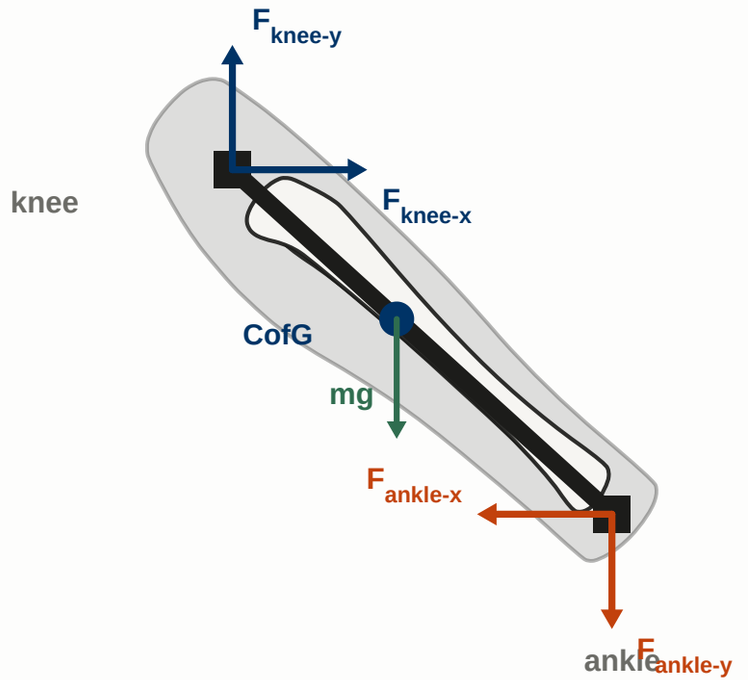
$$F_{\text{knee-x}} = 10.41\ \text{N}$$

$$\Sigma F_y = F_{\text{knee-y}} - F_{\text{ankle-y}} + mg = ma_{y(\text{leg})}$$

$$F_{\text{knee-y}} - 3.70\ \text{N} + (3.72)(-9.81) = (3.72\ \text{kg})(-4.21\ \text{m/s}^2)$$

$$F_{\text{knee-y}} = -15.66\ \text{N} + 3.70\ \text{N} + 36.49\ \text{N}$$

$$F_{\text{knee-y}} = 24.53\ \text{N}$$



The muscle force acting at the knee · the ankle reaction force

Inverse Dynamics Analysis Example

The leg, the moment from the knee forces

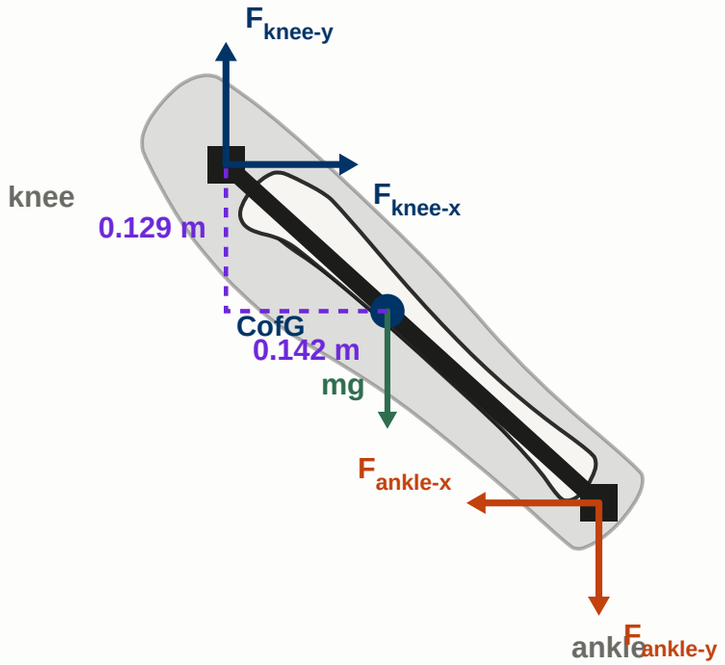
First solve for the moment caused by the forces at the **knee** about the CofG. *x* will cause a negative moment, *y* will cause a positive one.

$$M_{\text{Fknee/CG}} = -F_{\text{knee-x}}d_{\text{Px}} + F_{\text{knee-y}}d_{\text{Py}}$$

$$M_{\text{Fknee/CG}} = -(10.41\ \text{N})(0.129\ \text{m}) + (24.53\ \text{N})(0.142\ \text{m})$$

$$M_{\text{Fknee/CG}} = -1.34\ \text{N·m} + 3.48\ \text{N·m}$$

$$M_{\text{Fknee/CG}} = 2.14\ \text{N·m}$$



0.129 m is the **vertical** offset and pairs with the horizontal force; 0.142 m is the horizontal one.

Inverse Dynamics Analysis Example

The leg, the moment from the ankle forces

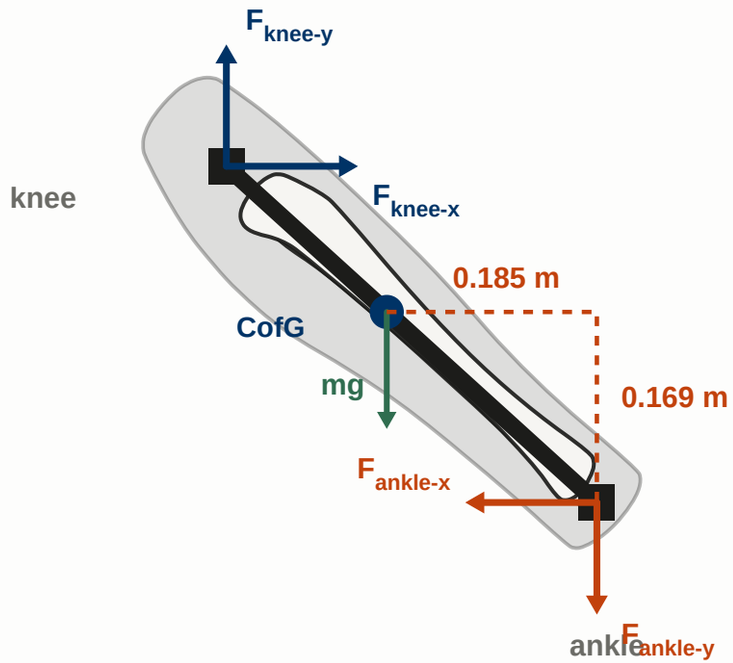
Then solve for the moment caused by the **ankle** forces about the CofG. *x* will cause a negative moment and *y* will cause a positive one.

$$M_{\text{Fankle/CG}} = -F_{\text{ankle-x}}d_{\text{Px}} + F_{\text{ankle-y}}d_{\text{Py}}$$

$$M_{\text{Fankle/CG}} = -(10.52\ \text{N})(0.169\ \text{m}) + (3.70\ \text{N})(0.185\ \text{m})$$

$$M_{\text{Fankle/CG}} = -1.78\ \text{N·m} + 0.68\ \text{N·m}$$

$$M_{\text{Fankle/CG}} = -1.09\ \text{N·m}$$



The same two rules, the other end of the segment, and a longer pair of arms.

Inverse Dynamics Analysis Example

The leg, the knee moment

Now solve for all the moments.

$$M_{\text{knee}} + M_{\text{Fknee/CG}} + M_{\text{Fankle/CG}} - M_{\text{ankle}} = I\alpha$$

$$M_{\text{knee}} = I\alpha - M_{\text{Fknee/CG}} - M_{\text{Fankle/CG}} + M_{\text{ankle}}$$

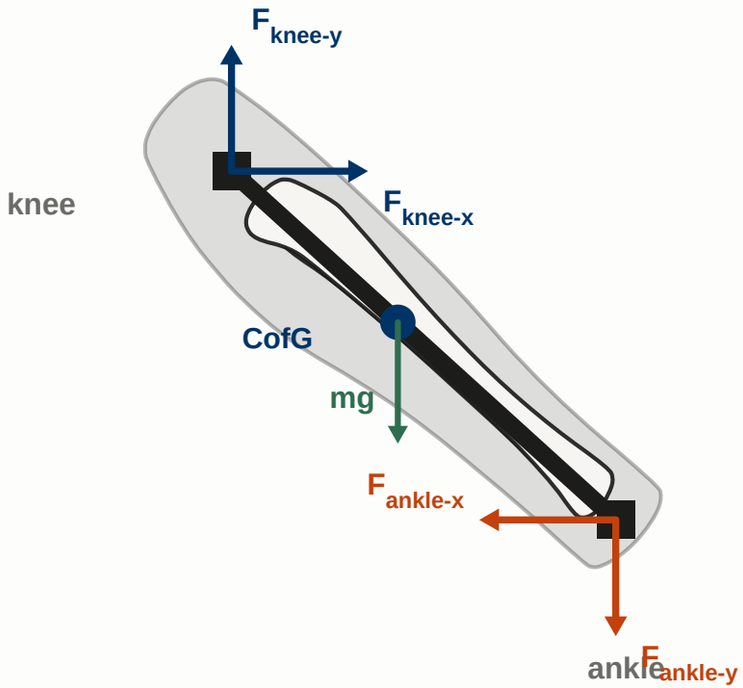
$$M_{\text{knee}} = (0.0642\ \text{kg·m}^2)(36.9\ \text{rad/s}^2) - 2.14\ \text{N·m} + 1.09\ \text{N·m} + 0.39\ \text{N·m}$$

$$M_{\text{knee}} = 2.37\ \text{N·m} - 2.14\ \text{N·m} + 1.09\ \text{N·m} + 0.39\ \text{N·m}$$

$$M_{\text{knee}} = 1.72\ \text{N·m}$$

This positive torque denotes a knee **extensor** moment.

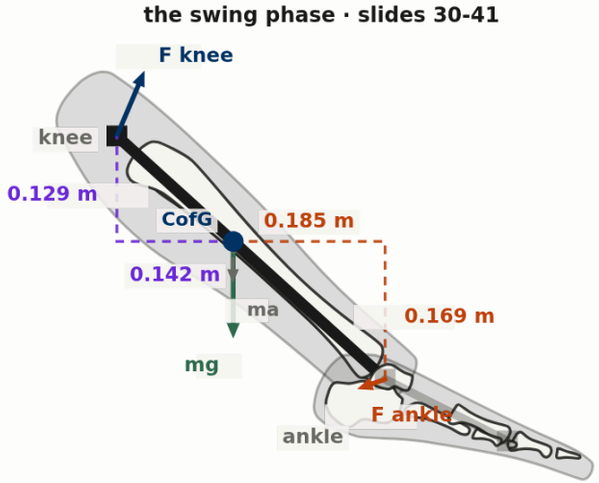
> Really just remember that what we solved for at the ankle will affect each limb **up the chain**.



Four terms, and they add to *Iα* — the four moments from earlier in the lecture, with real numbers in them.

Inverse Dynamics Analysis Example

The same example, and the same method in stance



4. the shank, ΣM = I·α

the knee force about the CoM: 2.141 N·m

the ankle force about the CoM: −1.094 N·m

and the ankle moment, reversed: −0.39 N·m

M(knee) = I·α − (2.14) − (−1.09) − (−0.39) = 1.72 N·m

M(F knee/CG)	the slide too
M(F ankle/CG)	the slide too
M(knee)	the slide too

Four steps, two segments, and the whole of the worked example. Solve the foot, reverse what you found, and hand it to the leg.

How much ground force — **100 %**

The swing example | Example 2, in stance

Foot: forces | Foot: moment | Shank: forces | **Shank: moment**

The same four steps, twice — and then a third thing the slides cannot do. Put it on **Example 2** and wind the ground force down from 100 % to nothing. The ankle moment goes from −76 N·m to almost zero and back, in proportion, with every other input held still. That is the answer to **where does an ankle moment come from**: in stance it is the ground reaction force and very little else.

On a real stride

The same method, run on measured data, and checked against somebody else’s answer.

On Real Data

The recording, and the table it is solved with

One stride of overground walking. A woman, 55 kg, 1.61 m, at 1.14 m/s. Markers at 60 Hz, the ground reaction force and the centre of pressure at 960 Hz, the plate under the right foot for the first 60 % of the cycle.

The authors **published their own joint moments** for this stride, from their own three-dimensional model — so we can do it from scratch and then mark our own work against somebody who was not trying to agree with us.

Lencioni *et al.* (2019), *Scientific Data* **6**:309, CC BY 4.0.

Dempster's table, as Winter tabulates it. The two blue columns are the m and the I in $\Sigma F = ma$ and $\Sigma M = I\alpha$, for all sixty-nine frames.

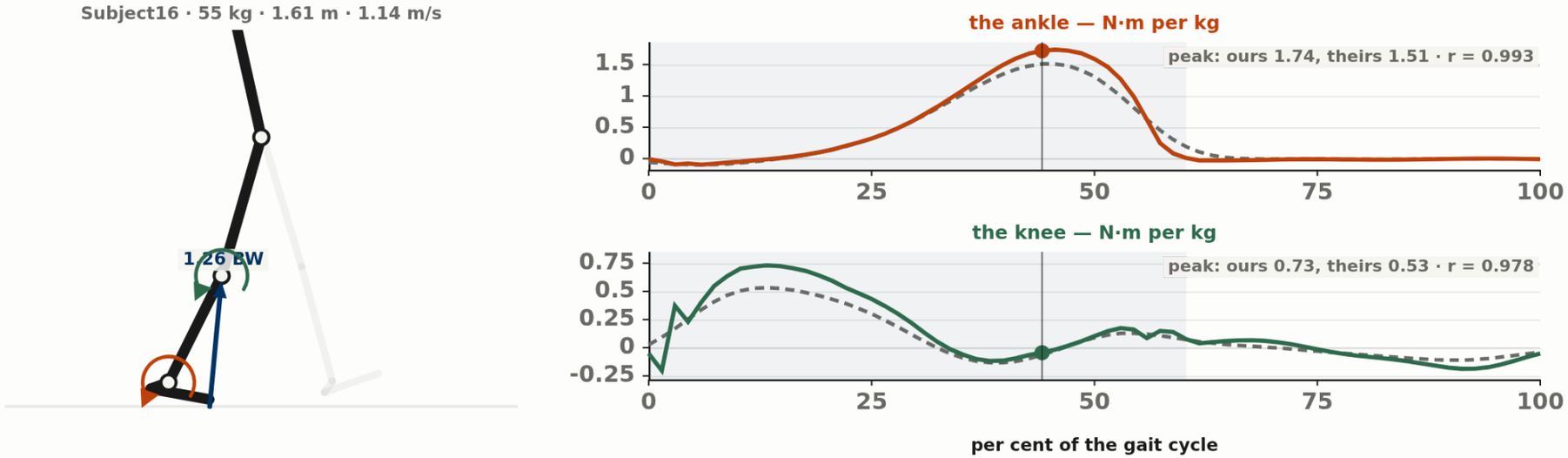
Segment	**mass** / body mass	**CoM** from proximal	***k*** / length	**length** measured	***m*** kg	***I*** kg·m²
Foot	0.0145	0.500	0.475	0.145 m	**0.797**	**0.00381**
Leg	0.0465	0.433	0.302	0.385 m	**2.558**	**0.03450**
Thigh	0.1000	0.433	0.323	0.392 m	**5.500**	**0.08817**

m = (mass fraction)(55 kg); $I = m(kL)^2$, with L off this subject's own markers. The foot's **0.797 kg** and **0.00381 kg·m²** are the two numbers the free body on slide 29 has been built from all along.

On Real Data

The same method, run for real

The chain from slide 18, the four moments from slide 25, all sixty-nine frames, *m* and *I* off Dempster's table.



Where in the stride — 44 %

Moments | Forces | **▶ Walk**

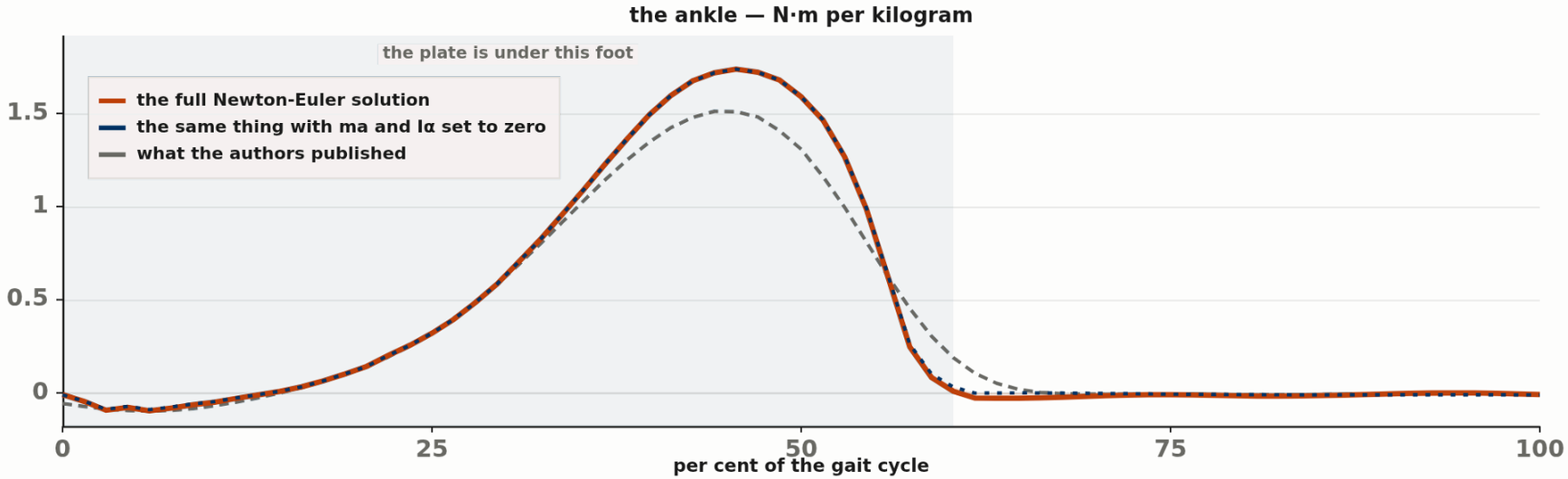
Both | What we worked out | What they published

One stride, two answers. The shapes match — r = 0.99 at the ankle — and both peaks land on the same instant; ours run 15 % high at the ankle and 37 % at the knee. **Nothing in the arithmetic is wrong: the gap is the model**, and the next slide shows the dynamic terms are not the cause of it.

On Real Data

When do you actually need the dynamics?

Slide 1 promised the only edit was zeros becoming *ma* and *Iα*. Here is that promise measured.



the dynamics are worth 0.06 % here

Over stance the two solutions differ by at most 0.023 N·m/kg. At the push-off peak the full solution gives 1.738 and statics gives 1.737 — a difference of 0.06 %. The foot weighs 0.8 kg and is barely accelerating; the ground is pushing with 678 N a hand's breadth from the ankle. Nothing else could matter.

The ankle | The knee

This is the figure lecture 19 asked for. Last week the walking ankle was solved as a **statics** problem and came out 15 % above the published answer, and the obvious suspect was the missing ma and Iα. It was not them. Setting every dynamic term to zero moves the ankle's push-off peak by six hundredths of one per cent. The 15 % is the model — whose anthropometric table, and where you decide the joint is. At the knee in swing it is the other way round entirely, and that is the case worth remembering: **no external force, a heavy segment, and dynamics is the whole of the answer**.

On Real Data

A different movement, the same three equations

Walking is the easy case: slow, cyclic, one foot on the plate at a time. So here is the method on something else entirely — a **countermovement jump**, measured in this building.

A 60.7 kg subject, 83 markers at 90 Hz, two force plates at 450 Hz with **one foot on each**, so the two records together are the whole body. He dips, drives up, is in the air for **0.456 s** — a 25.5 cm jump — and lands.

Nothing in the method changes. The right leg is solved foot, then shank, then thigh, at every one of the ninety-one frames.

Two things had to be fixed before any of it meant anything, and neither is in the paper.

1. The plate runs 0.1333 s behind the markers. Different in every trial of this dataset. Found by cross-correlating the marker centre of mass against (F_z − BW)/m: r = 0.997 at twelve frames, 0.4 at zero.

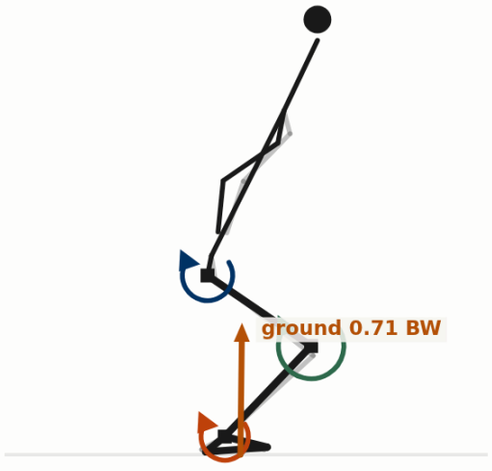
2. The centre of pressure is rotated ninety degrees. The file adds the plate centre on the wrong axes. Put back, it lands under each foot and walks heel to toe through the push-off — which is how you know it is right.

An inverse dynamics answer is only as good as the registration underneath it, and the registration is never in the data.

On Real Data

The jump, joint by joint

from the side



at 0.45 s — driving up

1.31 body weights, both feet together

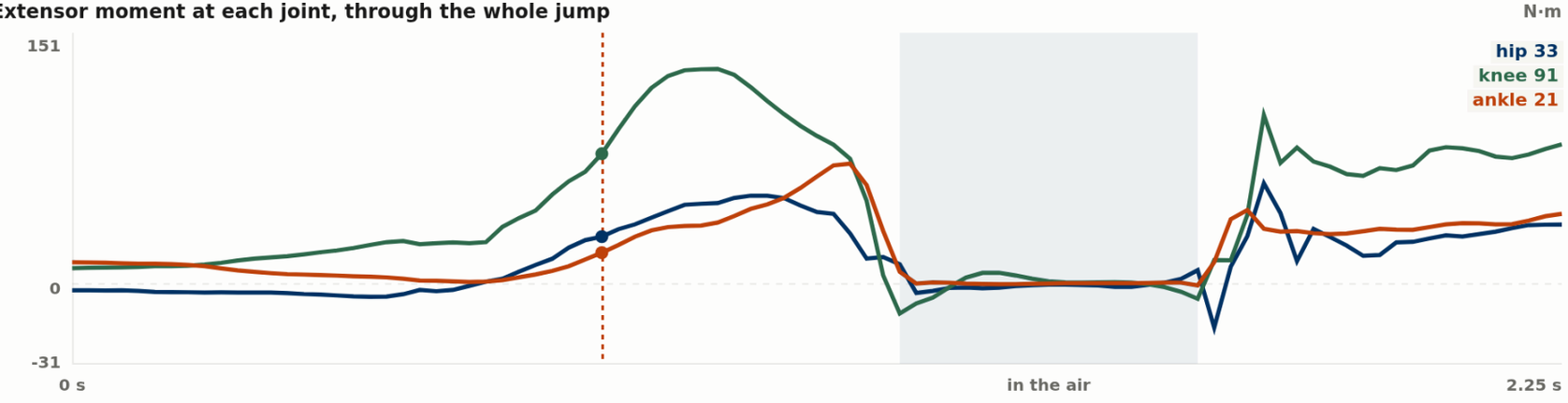
the ankle, knee and hip moments

21 · 91 · 33 N·m

the force the joints carry

412 · 383 · 308 N

Extensor moment at each joint, through the whole jump



Where in the jump — **0.45 s**

Moments | Forces | **From the side** | Turn it round | **▶ Jump**

One jump, the same three equations, ninety-one times. Watch the order the joints peak in: **knee first, then hip, then ankle**, over about two tenths of a second. Turn it round and the ground force barely leaves the sagittal plane — a third of a degree — which is the licence to work in two dimensions here.

On Real Data

What the jump shows that the walk did not

- **The joints peak in order.** Knee at 151 N·m, 0.28 s before take-off; hip at 62, 50 ms later; ankle at 84, with 0.07 s to go. Proximal to distal, and nobody put it there — it comes out of the three equations.
- **The size is different, the method is not.** 2.48 N·m/kg at the knee against 0.73 in the walk.
- **Landing is not the mirror of take-off.** The knee comes back to 118 N·m but the hip goes to 70, higher than it ever reached on the way up.

And a check the walk could not give us. The leg chain never touches a marker above the hip. So take the hip force it produces, double it for two legs, and compare it against what the trunk, head and arms independently need:

$$r = 0.992 \cdot \text{peaks } 979 \text{ N against } 901 \text{ N}$$

The gap is 8 % of peak, vertically. Horizontally the forces are under 100 N and the agreement is poor — which is the honest half of the result, and the reason the slide says *vertically*.

Dynamic Analysis

What to take away

- Dynamic analysis is static analysis with *ma* and *Iα* on the right. The diagram, the table and the conventions are unchanged.
- Three equations means three unknowns, which is why you start at the segment touching the ground and work upward.
- Four moments act on every segment, and they add to *I*α: one at each end, and one about the centre of mass from each end's force.
- Both the force **and** the moment reverse at every joint on the way up.
- Take moments about each segment's own centre of mass, and pair each force with the arm **perpendicular** to it.
- A joint moment is a **calculation**, not a measurement, and it is only as good as the anatomy you fed it.

Dynamic Analysis

Where this goes next

Multiply a joint moment by the joint's angular velocity and you get **joint power** — how fast that joint is putting energy into the body or taking it out. That one multiplication turns this lecture into the standard clinical description of walking.

And the honest limitation remains: inverse dynamics tells you what the joint did, never which muscle did it. Getting from a net moment to individual muscle forces needs either a measurement we cannot make, or a forward model and an assumption about what the nervous system is trying to optimise.

All done.